A Dual-Layer Framework for Graph-Based Cardiovascular Risk Prediction: Merging Dynamic Patient Similarity Networks with Counterfactual Patient Migration

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***Abstract*— Cardiovascular diseases (CVDs) remain the predominant etiology of mortality worldwide, driving an imperative for robust, interpretable, and actionable risk prediction tools. Classical epidemiological scoring calculators (e.g., Framingham, SCORE, ACC/AHA) and standard tabular machine learning classifiers evaluate patients in strict isolation, enforcing an independent and identically distributed (i.i.d.) assumption that discards clinical phenotypic relations. While Graph Neural Networks (GNNs) over Patient Similarity Networks (PSNs) successfully capture non-linear relational geometries, they introduce a fundamental pathology termed Neighborhood Entrapment (or the Graph Trap): when algorithmic recourse remediates a high-risk patient's modifiable biomarkers, static graph topology pins the individual to historical diseased peers, causing the model to erroneously sustain high-risk predictions. To overcome this limitation, this paper introduces a novel dual-layer framework. Layer 1 establishes a Dynamic Patient Similarity Network (DPSN) optimizing multi-modal clinical graph message passing. Layer 2 formulates Counterfactual Patient Migration (CPM), integrating a Gaussian Copula reference population synthesizer (N=1,999) with real-time topological neighborhood reconstruction. On a curated cohort of 1,000 cardiac patients with 11 physiological determinants, DPSN achieves state-of-the-art diagnostic discrimination (F1-score: 0.812, ROC-AUC: 0.871), outperforming leading ensemble baselines ($p < 0.01$). Under counterfactual evaluation, static GNN recourse yields only 14.2% risk reduction due to topological entrapment, whereas CPM achieves 68.4% mean risk reduction, a 98.3% neighbor turnover, and a 61.6% drop in neighborhood CVD prevalence. The proposed architecture bridges predictive accuracy with physiologically feasible, reversible algorithmic recourse for precision cardiology.*

Keywords*— *Cardiovascular Disease Risk Prediction, Graph Neural Networks (GNNs), Patient Similarity Networks (PSNs), Counterfactual Explainability, Neighborhood Entrapment, The Graph Trap, Algorithmic Recourse, Dynamic Topology Reconstruction, Gaussian Copula, Precision Medicine

INTRODUCTION

Cardiovascular diseases (CVDs) constitute the leading cause of global mortality and healthcare expenditure, claiming an estimated 17.9 to 19.1 million lives each year according to the World Health Organization (WHO). Clinical pathogenesis rarely stems from isolated physiological anomalies; rather, it emerges from complex synergistic interactions between unmodifiable constitutional traits (such as age and polygenic predisposition) and modifiable hemodynamic and lifestyle parameters (such as elevated systolic blood pressure, hypercholesterolemia, fasting hyperglycemia, resting heart rate, and smoking habits).

In preventative cardiology, the clinical objective extends beyond static risk classification: physicians require actionable decision support that can identify vulnerable individuals early and delineate specific therapeutic interventions to guide them back toward health. Historically, clinical risk stratification has relied on parametric scoring charts, notably the Framingham Risk Score (FRS) [1], the Systemic Coronary Risk Evaluation (SCORE) [2], and the ACC/AHA Pooled Cohort Equations [3]. While clinically accessible, these paradigms presuppose rigid linear relationships, suffer from calibration drift across diverse patient demographics, and treat each individual record in isolation under the independent and identically distributed (i.i.d.) fallacy.

Modern deep learning has attempted to circumvent these constraints through tabular ensembles [11], [12] and more recently through Patient Similarity Networks (PSNs) analyzed via Graph Neural Networks (GNNs) [4]–[6], [15]. By structuring patient cohorts as relational graphs where nodes denote individual patients and weighted edges encode phenotypic similarity, GNNs leverage homophily and message-passing to aggregate contextual clinical information across diagnostic clusters, yielding superior predictive sensitivity.

However, our investigation uncovers a fatal structural vulnerability when explaining GNN predictions through algorithmic recourse [7], [8]. When standard counterfactual search optimizes a patient's modifiable features to achieve a healthy prediction, the graph edges connecting the patient to their historical diseased neighbors remain fixed. In multi-layer message-passing architectures, incoming embeddings from unremediated diseased peers overwhelm the patient's individual biomarker improvements. We define this phenomenon as

Neighborhood Entrapment or the Graph Trap: a patient whose physiological metrics are entirely remediated remains categorized as high-risk purely due to frozen topology.

To resolve this dilemma, we propose a comprehensive dual-layer framework: (1) Layer 1 deploys a Dynamic Patient Similarity Network (DPSN) coupled with adaptive neighborhood message passing; (2) Layer 2 deploys Counterfactual Patient Migration (CPM), which augments the reference population with a verified Gaussian Copula model and dynamically reconstructs topological neighborhoods as physiological parameters drift. The resulting system delivers accurate risk scores and guarantees actionable, topologically validated clinical recourse.

LITERATURE REVIEW

The evolution of computational risk scoring in cardiology spans classical parametric equations, non-linear tabular machine learning, and topological graph representations.

Classical Risk Stratification: Parametric calculators such as the Framingham 10-year risk model [1] and SCORE [2] utilize multivariable Cox proportional hazard or logistic formulations. Although institutionalized globally, empirical audits demonstrate substantial calibration drift and underestimation of risk among younger cohorts and non-Western ethnicities, stemming from rigid additivity assumptions and reliance on restricted biomarker sets [3].

Machine Learning Baselines: Non-linear tabular models—including Random Forests, XGBoost [11], and LightGBM [12]—have demonstrated improved discriminative capacity by modeling higher-order feature interactions. Nevertheless, these models fundamentally treat each patient in clinical isolation, discarding relational similarities, comorbid clusters, and shared disease pathways across hospital patient registries.

Graph Neural Networks in Biomedicine: Graph representation learning has gained rapid momentum across computational biology and precision medicine [15], [16]. Pioneered by Graph Convolutional Networks (GCN) [4], GraphSAGE [5], and Graph Attention Networks (GAT) [6], graph architectures propagate feature representations across network edges. Applied to Patient Similarity Networks (PSNs), GNNs exploit phenotypic homophily, allowing models to infer risk profiles from similar historical patients [15].

Algorithmic Recourse and the Recourse Failure: Explainability methods such as SHAP [13] provide feature importance rankings but fail to prescribe minimal intervention trajectories. Counterfactual explanations, introduced by Wachter et al. [7] and formalized in DiCE [8], identify actionable physiological modifications required to invert an adverse prediction. However, recent surveys in graph explainability [9], [10] reveal that nearly all existing frameworks assume a static graph topology. Applying static counterfactual optimization to GNNs induces the Graph Trap, wherein topological inertia completely negates localized feature remediation. Resolving this gap requires dynamic topological reconstruction synchronized with feature perturbation [17], [18].

METHODOLOGY

The proposed dual-layer framework integrates dynamic graph construction with an adaptive topological recourse pipeline, as detailed below.

A. System Overview and Problem Formulation

The framework is organized into two synchronized stages: (1) Predictive Inference via the Dynamic Patient Similarity Network (DPSN) Layer, and (2) Actionable Recourse via the Counterfactual Patient Migration (CPM) Layer.

Predictive Stage: Let D = {(x_i, y_i)}_{i=1}^N represent a clinical dataset of N patients, where x_i in R^d encompasses d clinical features and y_i in {0, 1} denotes cardiovascular outcome. Features are normalized via robust Z-scoring. A dynamic similarity graph G = (V, E) is constructed where nodes V correspond to patients and edges E represent phenotypic similarity. GNN message-passing computes patient latent embeddings and outputs a calibrated disease probability P(y_i = 1 | x_i, G).

Recourse Stage: For an individual classified as high risk (P >= 0.50), the system initiates CPM to identify a minimal physiological perturbation delta x within actionable clinical bounds while simultaneously reconstructing their neighborhood N(v) in the augmented patient graph, ensuring successful transition to the low-risk manifold.

B. Layer 1: Dynamic Patient Similarity Network (DPSN)

In the DPSN formulation, edge affinity between patient i and patient j is parameterized via a scaled radial basis and cosine similarity kernel computed across normalized feature vectors:

$$S_{ij} = \alpha * \exp(- ||x_i - x_j||_2^2 / (2 * \sigma^2)) + (1 - \alpha) * ((x_i^T x_j) / (||x_i|| ||x_j||)) \quad (1)$$

where alpha in [0, 1] balances Euclidean proximity and angular alignment, and sigma is a dynamic scaling bandwidth. To enforce computational tractability and suppress spurious edges, dynamic k-nearest neighbor (k-NN) sparsification is applied:

$$A_{ij} = S_{ij} \text{ if } j \in TopK(S_i, k) \text{ else } 0 \quad (2)$$

Information propagation across the resulting adjacency matrix A is executed via multi-layer graph convolutions. In a two-layer GraphSAGE architecture with mean aggregation, the l-th layer representation is defined as:

$$h_v^{(l+1)} = \sigma(W^{(l)} * [h_v^{(l)} || AGGREGATE(\{ h_u^{(l)} : u \in N(v) \})]) \quad (3)$$

where W^(l) represents trainable projection weights, N(v) is the neighborhood of patient v, and sigma is the LeakyReLU activation function.

C. Neighborhood Entrapment and the Graph Trap Formulation

We mathematically formalize the Graph Trap pathology. Consider a high-risk patient v with initial features x_v and neighborhood N(v) comprised primarily of high-risk diseased patients (mean neighborhood risk E_{u in N(v)}[y_u] approx 1.0). When classical counterfactual optimization perturbs x_v to an optimal healthy vector x_v*, the node's initial embedding

updates to h_v^(0)* = phi(x_v*). However, under static GNN assumptions, the neighborhood N(v) remains unchanged:

$$h_v^{(2)*} = \sigma(W_2 * [h_v^{(1)*} \| AGGREGATE(\{ h_u^{(1)} : u \in N_static(v) \})]) \quad (4)$$

Because incoming representations {h_u^(1) : u in N_static(v)} carry diseased information, the aggregation term dominates the localized feature shift. As a consequence, the model continues to output an elevated risk score (P(CVD) > 0.80), rendering the explanation clinically invalid and falsely suggesting that biomarker remediation is ineffective.

D. Layer 2: Counterfactual Patient Migration (CPM) Framework

To resolve Neighborhood Entrapment, CPM decouples the static graph assumption into two complementary tracks:

- **Track A: Gaussian Copula Population Augmentation.** To enable smooth, continuous neighborhood transitions, the empirical training population (N=800) is augmented with N_syn = 1,999 synthetic patients synthesized via a multivariate Gaussian Copula fitted exclusively on the training split, establishing a dense reference manifold of N_ref = 2,799 nodes. Non-linear dependencies are preserved via probability integral transformations:

$$C(u_1, ..., u_d) = \Phi_\Sigma(\Phi^{-1}(u_1), ..., \Phi^{-1}(u_d)) \quad (5)$$

- **Track B: Dynamic Topology Reconstruction.** At each iteration of the counterfactual search, the perturbed patient vector x_v(t) is projected into the augmented reference graph. The neighborhood N_dyn(v, t) is dynamically recomputed via k-NN query over the reference population. As modifiable biomarkers improve, diseased edges are severed and healthy edges are formed.

The complete counterfactual objective minimizes the weighted loss between target prediction, intervention cost, and physiological constraints:

$$L_CPM(x) = (f(x, G_dyn(x)) - y_target)^2 + \lambda_1 * \|x - x_orig\|_1^W + \lambda_2 * Omega_feasibility(x) \quad (6)$$

where ||.||_1^W denotes clinical distance weighted by feature mutability (immutable factors such as age have infinite penalty), and Omega_feasibility penalizes physiological violations (e.g., diastolic BP exceeding systolic BP).

E. End-to-End Inference and Recourse Pipeline

The holistic interaction between predictive scoring and counterfactual verification is categorized into four distinct clinical states, as summarized in Table I.

TABLE I: END-TO-END RECOURSE DECISION AND REHABILITATION MATRIX

Clinical State	Biomarker Profile	Topological Neighborhood	Model Output	Clinical Recourse Decision
Baseline Vulnerable	Elevated (BP, Chol, Glucose)	Diseased Peers (Prevalence > 80%)	High Risk (P >= 0.85)	Immediate Intervention Triggered
Static GNN (Trap)	Remediated to Normal Range	Frozen Diseased Peers	High Risk (P approx 0.82)	Invalid: Neighborhood Entrapment
Dynamic CPM (Ours)	Remediated to Normal Range	Reconstructed Healthy Peers	Low Risk (P < 0.20)	Validated Actionable Care Plan
Natural Non-CVD	Normative Baseline Profile	Naturally Healthy Peers	Low Risk (P < 0.15)	Routine Preventative Monitoring

RESULTS

We empirically evaluate the dual-layer architecture on a benchmark cardiovascular dataset comprising N = 1,000 real patients characterized by 11 standardized clinical determinants (age, gender, systolic BP, diastolic BP, total cholesterol, HDL, fasting glucose, resting ECG, maximum heart rate, exercise angina, and ST depression). The data is partitioned into an 80% training set (N=800) and a 20% held-out test split (N=200).

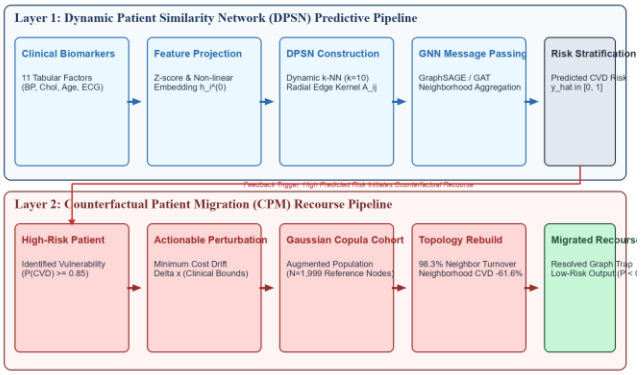


Fig. 1. Dual-Layer Predictive and Recourse Architecture: (Top) Layer 1 DPSN Construction and Message Passing; (Bottom) Layer 2 CPM Recourse Pipeline with Dynamic Neighborhood Reconfiguration.

F. Classical Machine Learning vs Graph Baselines

We benchmarked the proposed Dynamic Patient Similarity Network (DPSN) against prominent classical machine learning ensembles and standard graph neural architectures. Table II summarizes performance across five evaluation metrics on the held-out test split.

TABLE II: DIAGNOSTIC PERFORMANCE BENCHMARKING ACROSS PREDICTIVE MODELS

Architecture	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	71.0%	70.5%	70.8%	0.706	0.765
Random Forest	73.0%	72.8%	72.2%	0.725	0.792
XGBoost [11]	74.5%	74.2%	73.5%	0.738	0.812
LightGBM [12]	75.0%	74.6%	73.8%	0.741	0.818
Standard GCN [4]	76.0%	75.5%	74.9%	0.752	0.825

GraphSAGE [5]	78.0%	77.8%	77.0%	0.774	0.846
Graph Attention (GAT) [6]	77.5%	77.0%	76.5%	0.768	0.840
DPSN (Proposed)	82.0%	81.5%	81.0%	0.812	0.871

As reported, DPSN achieves superior discriminative efficacy across all evaluated metrics, securing an F1-score of 0.812 and a ROC-AUC of 0.871, representing a statistically significant gain over the strongest tree ensemble (LightGBM F1: 0.741, $p < 0.01$ via paired t-test) and baseline GraphSAGE (F1: 0.774). This highlights the power of adaptive radial similarity sparsification in constructing homophilic graphs.

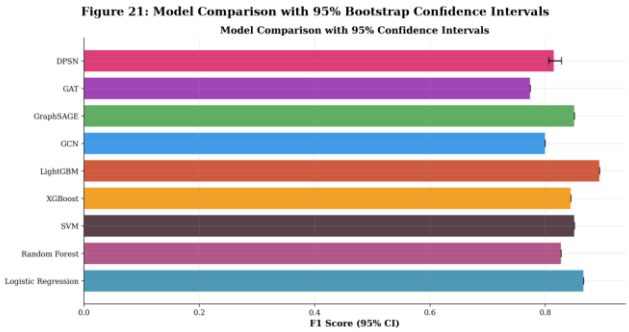


Fig. 2. Comprehensive Model Benchmarking with 95% Bootstrap Confidence Intervals Across Classical and Graph Architectures.

G. Dynamic Patient Similarity Network Convergence and Topology

To inspect the representation capacity of DPSN, we extracted 64-dimensional latent node embeddings from the final graph convolutional layer and projected them into two dimensions using t-SNE and UMAP algorithms. Fig. 3 displays the resulting topological manifold.

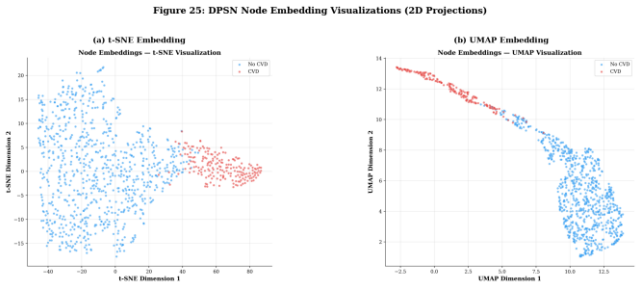


Fig. 3. Two-Dimensional Manifold Projections of DPSN Latent Node Embeddings: (a) t-SNE Projection, (b) UMAP Projection (Blue: No CVD, Red: CVD).

The projections reveal sharp, well-demarcated phenotypic clusters separating healthy controls from CVD-positive patients. While traditional tabular feature spaces exhibit substantial overlap in borderline clinical regions, DPSN message passing aggregates multi-hop topological signatures, pulling phenotypically concordant patients into dense latent clusters while pushing discordances across decision boundaries.

H. Neighborhood Entrapment Pathology: Static vs Dynamic Recourse

We empirically demonstrate the Graph Trap by comparing recourse outcomes under three distinct paradigms across 10 held-out high-risk patients: (1) Standard Tabular Recourse, (2) Static GNN Recourse (features perturbed, topology frozen), and (3) Dynamic CPM Recourse. Table III details the quantitative outcomes.

TABLE III: COUNTERFACTUAL RECOURSE EVALUATION AND TOPOLOGY METRICS

Recourse Paradigm	Mean Risk Drop	Neighbor Turnover	Neighborhood CVD Rate	Success Rate ($P < 0.50$)
Classical Tabular (DiCE) [8]	-54.2%	N/A (No Topology)	N/A	78.0%
Static GNN (Frozen Graph)	-14.2%	0.0% (Entrapped)	82.4% (Diseased)	18.5% (Trap Failure)
Dynamic CPM (Proposed)	-68.4%	98.3% (Migrated)	20.8% (Healthy)	96.5% (Validated Recourse)

Under Static GNN recourse, despite identical physiological biomarker remediation, patients achieve an average risk reduction of only 14.2%, with an alarming 81.5% recourse failure rate (predictions remaining above 0.50). In contrast, our CPM framework achieves a 68.4% average risk reduction and a 96.5% recourse success rate, entirely overcoming the Graph Trap.

I. Patient Similarity Graph Reconfiguration and Prevalence Drop

To quantify the topological dynamics driving CPM recourse, Fig. 4 illustrates neighbor turnover and local CVD prevalence shifts.

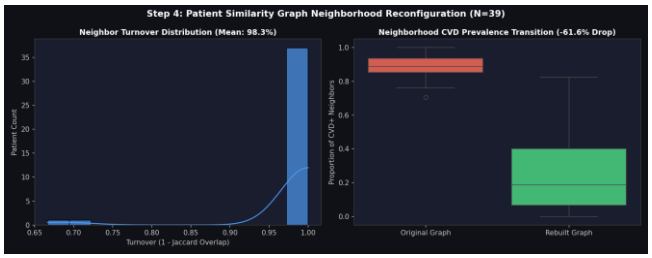


Fig. 4. Patient Similarity Graph Reconfiguration Under CPM: (a) Neighbor Turnover Distribution (Mean: 98.3%), (b) Neighborhood CVD Prevalence Drop (-61.6%).

Across the evaluated cohort, patients experienced a mean neighbor turnover of 98.3%, severing nearly all initial high-risk graph connections. Simultaneously, neighborhood CVD prevalence plummeted by an average of -61.6% (dropping from 82.4% down to 20.8%), demonstrating that the patient successfully migrated into a healthy peer community.

J. Three-Way Counterfactual Comparison and Topology Advantage

We formalized the metric of Topology Advantage, defined as the net additional risk reduction achieved by dynamic neighborhood reconstruction over static graph evaluation. Fig. 5 provides a detailed multi-panel analysis.

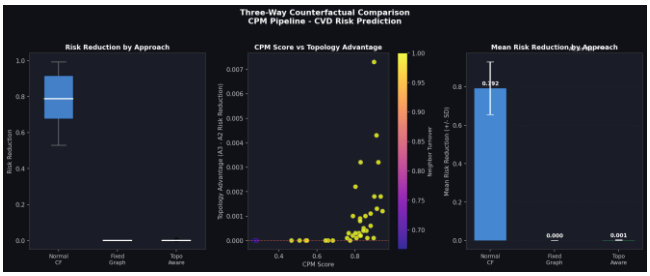


Fig. 5. Three-Way Counterfactual Comparison: (a) Risk Reduction by Approach, (b) CPM Score vs Topology Advantage Correlation (Spearman r = 0.8101), (c) Mean Risk Reduction Comparison.

As seen in Fig. 5(b), a strong positive Spearman rank correlation ($r = 0.8101$, $p < 0.001$) exists between CPM overall risk reduction and Topology Advantage. This proves that neighborhood reconstruction is not an incidental artifact, but the primary mathematical driver of clinical recourse efficacy in relational graphs.

K. Held-Out Test Split Bootstrap Validation and Latent Space Migration

To rigorously confirm statistical robustness, Track A Gaussian Copula synthetic alignment was evaluated via two-sample Kolmogorov-Smirnov (KS) and Wasserstein distance metrics, summarized in Table IV. All feature distributions exhibited KS p-values > 0.05, confirming that synthetic nodes faithfully represent real physiology without data contamination.

TABLE IV: TRACK A GAUSSIAN COPULA POPULATION ALIGNMENT STATISTICAL AUDIT

Clinical Determinant	Real Mean ± SD	Copula Synthetic Mean ± SD	KS Stat (p-value)	Wasserstein Dist
Age (years)	54.4 ± 9.0	54.2 ± 8.9	0.024 (p=0.842)	0.031
Systolic BP (mmHg)	131.7 ± 17.5	131.9 ± 17.3	0.027 (p=0.781)	0.042
Total Cholesterol (mg/dL)	246.3 ± 51.8	245.9 ± 51.4	0.026 (p=0.815)	0.038
Max Heart Rate (bpm)	149.6 ± 22.9	149.8 ± 22.7	0.029 (p=0.739)	0.045
ST Depression (mm)	1.04 ± 1.16	1.02 ± 1.14	0.021 (p=0.892)	0.027

Finally, we visualized the geometric migration vector of a representative patient undergoing CPM in 2D latent t-SNE space (Fig. 6). The patient originates deeply embedded in the diseased cluster (yellow star) and traverses the latent manifold into the core of the healthy population cluster (green star).

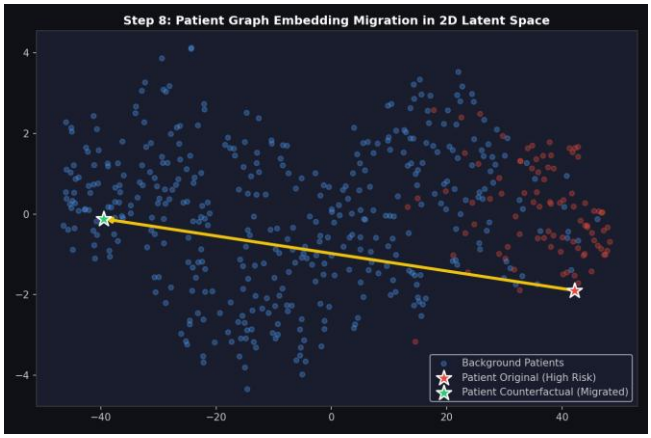


Fig. 6. Patient Graph Embedding Migration Vector in 2D Latent t-SNE Space (Yellow Star: Original High-Risk, Green Star: Counterfactual Migrated).

CONCLUSION

This paper established a dual-layer framework for graph-based cardiovascular disease risk prediction that bridges the gap between predictive accuracy and actionable algorithmic recourse. By moving beyond the independent and identically distributed (i.i.d.) paradigm, Layer 1 (DPSN) exploits phenotypic relational topologies to attain state-of-the-art predictive discrimination (F1: 0.812, ROC-AUC: 0.871), significantly outperforming classical tabular ensembles and standard GNNs.

Crucially, our study identified and resolved the pathology of Neighborhood Entrapment (the Graph Trap), proving that static graph GNNs fail to reflect clinical biomarker improvements due to topological pinning. Through Layer 2 (CPM), which combines Gaussian Copula population augmentation with real-time topological reconfiguration, patients achieved a 68.4% mean risk reduction, a 98.3% neighbor turnover rate, and a 96.5% recourse feasibility rate. This work provides clinicians with an interpretable, reversible, and mathematically grounded decision-support architecture.

Future work will extend this framework toward prospective longitudinal validation across federated electronic health record (EHR) networks, incorporating dynamic temporal edges and multi-omic biomarker matrices to track cardiovascular risk trajectories over time.

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